Project Title	**Aerial Object Classification & Detection**
Skills take away From This Project	• Deep Learning • Computer Vision • Image Classification & Object Detection • Python • TensorFlow/Keras or PyTorch • Data Preprocessing & Augmentation • YOLOv8 (Optional – Object Detection) • Model Evaluation • Streamlit Deployment
Domain	**Aerial Surveillance, Wildlife Monitoring, Security & Defense Applications**

📌 Problem Statement

This project aims to develop a deep learning-based solution that can **classify** aerial images into two categories — **Bird** or **Drone** — and optionally perform **object detection** to locate and label these objects in real-world scenes.

The solution will help in **security surveillance**, **wildlife protection**, and **airspace safety** where accurate identification between drones and birds is critical. The project involves building a **Custom CNN classification model**, leveraging **transfer learning**, and optionally implementing **YOLOv8** for real-time object detection. The final solution will be deployed using **Streamlit** for interactive use.

📌 Real-Time Business Use Cases

1. **Wildlife Protection**
   - Detect birds near wind farms or airports to prevent accidents.
2. **Security & Defense Surveillance**
   - Identify drones in restricted airspace for timely alerts.
3. **Airport Bird-Strike Prevention**
   - Monitor runway zones for bird activity.
4. **Environmental Research**
   - Track bird populations using aerial footage without misclassification.

📌 Project Workflow

1. Understand the Dataset

- Inspect dataset folder structure
- Check number of images per class
- Identify class imbalance
- Visualize sample images

---

2. Data Preprocessing

- Normalize pixel values to **[0, 1]**
- Resize images to a fixed size (**224×224** for classification)

- For **TensorFlow-based transfer learning** models, use the model-specific `preprocess_input` function.
- For **PyTorch-based pretrained models**, apply ImageNet normalization using `torchvision.transforms.Normalize(mean, std)` as per the model's training configuration.

---

3. Data Augmentation

- Apply transformations: rotation, flipping, zoom, brightness, cropping

---

4. Model Building (Classification)

- **Custom CNN:** Conv layers, pooling, dropout, batch normalization, dense output layer
- **Transfer Learning:** Load models like ResNet50, MobileNet, EfficientNetB0 and fine-tune

---

5. Model Training

- Train both models
- Use EarlyStopping & ModelCheckpoint
- **Track metrics:** Accuracy, Precision, Recall, F1-score

---

6. Model Evaluation

- Evaluate test results with confusion matrix & classification report
- Plot accuracy/loss graphs

7. Model Comparison

- Compare accuracy, training time, and generalization performance
- Save the best performing model for Streamlit deployment

📌 Optional: Object Detection with YOLOv8

Steps:

1. Install YOLOv8.
2. Prepare dataset (images and YOLOv8-format `.txt` labels — already done).
3. Create a `data.yaml` configuration file for YOLOv8.
4. Train the YOLOv8 model.
5. Validate the trained model.
6. Run inference on test or new images.

📌 Streamlit Deployment

- Create a simple UI with image upload

- Display prediction (Bird / Drone) & confidence score
- (Optional) Show YOLOv8 detection results with bounding boxes

---

📌 Project Deliverables

1. Trained models (Custom CNN, Transfer Learning, YOLOv8 (optional))
2. Streamlit app for classification/detection
3. Scripts & notebooks for preprocessing, training, evaluation
4. Model comparison report
5. GitHub repository with documentation
6. Well-structured, commented code

🛠️ **Technical Tags**

`Computer Vision, Deep Learning, Image Classification, Object Detection, CNN, YOLOv8, Transfer Learning, Data Augmentation, Model Evaluation, Streamlit Deployment, Aerial Surveillance AI`

📌 Datasets

📌 Classification Dataset

- **Source:** classification_dataset
- **Task:** Image Classification (Binary: Bird / Drone)
- **Data Type:** RGB Images

- **Format:** `.jpg`

Structure

- **TRAIN set:**
  - `- bird: 1414 images`
  - `- drone: 1248 images`
- **VALID set:**
  - `- bird: 217 images`
  - `- drone: 225 images`
- **TEST set:**
  - `- bird: 121 images`
  - `- drone: 94 images`

📌 Object Detection Dataset (YOLOv8 Format)

- **Source :** object_detection_Dataset
- The dataset contains 3319 images with corresponding YOLOv8-format annotations (`.txt` files).
- Each annotation file contains bounding boxes in the format:

```
<class_id> <x_center> <y_center> <width> <height>
```

- Data split: Train (2662), Validation (442), Test (215).

⏳ Timeline

The project should be completed and submitted **within 10 days** from the date it is assigned.

Reference

Streamlit recording (English)	Special session for STREAMLIT(11/08/2024)
Streamlit Reference doc	Streamlit API reference
Project Live Evaluation	Project Live Evaluation
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Deep learning material	Deep_Learning-study_material.pdf

Created By	Verified By	Approved By
Nilofer Mubeen	Shadiya	Nehlath Harmain

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Saturday (3:30PM to 4:30PM)

Booking link : Project Doubt Session [DS/AIML]

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA
